

Multimedia Information Retrieval for Sports Analytics

FIFA World Cup 2022 Knowledge Graph

Information Retrieval Exam Project

Course: Information Retrieval | Academic Year: 2025/2026

Abstract and objectives

This project builds a source-grounded knowledge graph for retrieval over FIFA World Cup data. It cleans and audits the supplied tabular sources, models tournament, team, player, match, goal, appearance, venue and image facts as RDF triples, imports the graph into Neo4j, and evaluates read-only Cypher queries against source-derived answers. A Streamlit dashboard exposes graph exploration and saved retrieval examples. The optional language-model feature is not treated as retrieval evidence.

The objective is to connect facts that are separated across datasets—for example, retrieving a match, its venue and score, goal events and scorers, or a player's nationality, club and image features—through explicit graph relationships.

Dataset coverage

Source / category	Records used	Integration
Historical World Cup summary	22	Tournament outcomes and totals
FIFA 2022 team standings	32	Team tournament statistics
FIFA 2022 player statistics	814	Player profile and performance
FIFA 2022 all-player roster	830	Player name coverage
jfjelstul match data	64	2022 match facts and venues
jfjelstul goal events	172	Goal minute, scorer and team
jfjelstul player appearances	1995	Match roster and roles
FIFA player image dataset	831 folders / 41510 files	Representative image and descriptors

The current RDF build contains **47894** triples. The match, goal and appearance tables are filtered to the men's FIFA World Cup 2022 tournament identifier (WC-2022); the supplied historical summary covers 22 tournaments.

1. Data and preprocessing

The local input collection includes the historical results table, a separate 2022 team standings table, 2022 player statistics, an all-player name list and 41,510 JPG assets organised into player folders. The professor-required jfflstul/worldcup source additionally supplies the match, goal and player appearance CSVs. A fetch script pins a specific upstream commit so the added files can be reproduced instead of relying on an untracked manual download.

Cleaning and quality controls

CSV column names and string cells are whitespace-trimmed. Empty strings, dash placeholders and common null markers are normalized to missing values; numeric and percentage values are parsed into typed RDF literals. Original row counts are retained, and records with missing optional facts are not discarded. World Cup match, goal and appearance sources are filtered by tournament_id rather than by a loose year/name substring. Stable source identifiers are kept for match, goal and player records; image paths are stored relative to the image root.

A JSON data-quality audit accompanies each graph build, recording row counts, columns, duplicate rows and missing values by column. Source files are not modified. Missing optional values are omitted from RDF instead of being converted to misleading text such as “-”.

Missing-value observations

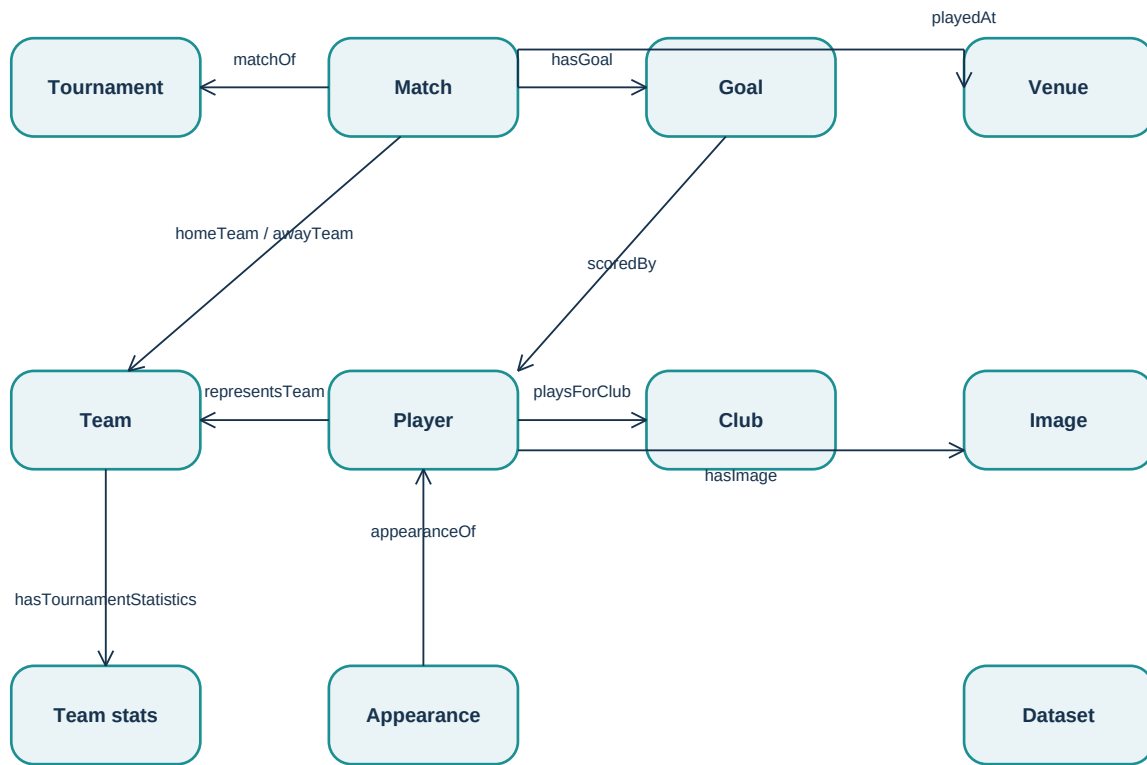
Player statistic	Missing rows	Treatment
Goals Scored	82	Source placeholder retained as missing; not assumed to mean zero.
Save Percentage	763	Role-specific field; missing values are not imputed for non-goalkeepers.
Clean Sheets	763	Role-specific field; missing values are not imputed for non-goalkeepers.
Appearances	7	Optional source field; no inferred value is added.

Reproducibility

Run `./run.sh` on macOS with Neo4j installed and configured in `.env`. The script installs declared Python dependencies, downloads any missing pinned match-source CSVs, rebuilds the Turtle graph, imports it, runs the benchmark suite, regenerates this report and starts Streamlit on port 8501. Importing with `--replace` replaces the selected Neo4j database; the importer without that option is additive.

2. Ontological schema

The global schema uses stable URI resources for entities and RDF literals for attribute values. RDF type assertions distinguish the core classes. Named relationships preserve the connections needed for multi-hop retrieval; `wasDerivedFrom` links entities and events to dataset resources for provenance.



Entities and events link to Dataset via `wasDerivedFrom`

Figure 1. Domain ontology (arrows are directed RDF predicates). Match-level facts are connected to teams, venues, tournaments and goal events. Players connect to team, club, appearance and image resources.

Representative predicates

Tournament: `hasChampion`, `hasRunnerUp`, `hostedIn`, `hasMatchCount`

Match: `matchOf`, `homeTeam`, `awayTeam`, `playedAt`, `hasHomeTeamGoals`, `hasAwayTeamGoals`

Goal: `scoredBy`, `forTeam`, `hasMinute`, `isPenalty`, `isOwnGoal`

Player: `hasNationality`, `hasPosition`, `playsForClub`, `representsTeam`, `hasImage`

Appearance: `appearanceOf`, `appearanceIn`, `playedFor`, `isStarter`, `isSubstitute`

Image: `hasFileName`, `hasImageCount`, `hasColorHistogramRgb8`

Provenance: `wasDerivedFrom`

3. Knowledge graph construction and retrieval

The graph builder produces RDF subject-predicate-object statements with rdflib and serializes them as Turtle. Neo4j stores URI resources and literal-value nodes; predicate local names become relationship types. Literal values retain their RDF datatype and language on their outgoing relationships, while resource nodes record their category. Imports are batched by predicate. The default import is additive; a full replacement requires the explicit `--replace` flag.

Example semantic queries

Champion of 2022:

```
MATCH (t:Resource)-[:hasChampion]->(v:Literal) WHERE t.uri ENDS WITH '/Tournament/2022'  
RETURN v.value
```

2022 matches with venue and score:

```
MATCH (m:Resource)-[:matchOf]->(t:Resource), (m)-[:playedAt]->(v:Resource),  
(m)-[:hasScore]->(s:Literal) WHERE t.uri ENDS WITH '/Tournament/2022' RETURN m.uri, v.uri,  
s.value
```

Goal events by player:

```
MATCH (m:Resource)-[:hasGoal]->(g:Resource), (m)-[:matchOf]->(t:Resource),  
(g)-[:scoredBy]->(p:Resource) WHERE t.uri ENDS WITH '/Tournament/2022' RETURN p.uri,  
count(g)
```

The Streamlit dashboard includes saved examples, direct Cypher retrieval, graph visualization and optional AI chat. Dashboard queries run in Neo4j read transactions. AI responses are an optional explanation layer and are not used for correctness or evaluation.

4. Multimedia feature extraction

The image tree contains thousands of files, so the reproducible graph uses one deterministically selected representative JPEG per player folder. Pillow records original width and height, relative file path and image count. It also resizes the representative to 64×64 RGB and computes a normalized 24-value color histogram (8 bins per RGB channel). The compact descriptor is stored as an RDF feature literal linked to the image resource. This is a transparent low-level visual descriptor, not a learned embedding or claim of face recognition.

Neo4j queries can retrieve image resources, their source counts and histogram descriptors. The method is suitable for metadata/color-based exploration; future work could add perceptual hashes or learned embeddings and explicit similarity ranking.

5. Performance evaluation

Neo4j read-only queries; one warm-up followed by repeated runs; median wall-clock latency reported.

Benchmark / source	Expected	Returned	Correct	Median ms
Historical FIFA 2022 champion World Cup summary	[{"champion": "Argentina"}]	[{"champion": "Argentina"}]	Yes	6.801
Argentina tournament points FIFA 2022 team standings	[{"points": 18}]	[{"points": 18}]	Yes	1.299
Player profile and club FIFA 2022 player statistics	[{"nationality": ["Argentina"], "position": ["FW"], "club": ["PSG"]}]	[{"club": ["PSG"], "nationality": ["Argentina"], "position": ["FW"]}]	Yes	0.745
FIFA 2022 match coverage jfelstul matches.csv	[{"matches": 64}]	[{"matches": 64}]	Yes	0.790
Lionel Messi goal events jfelstul goals.csv	[{"goals": 7}]	[{"goals": 7}]	Yes	0.688
Lionel Messi match appearances jfelstul player_appearances.csv	[{"appearances": 7}]	[{"appearances": 7}]	Yes	0.638
Player image feature coverage FIFA 2022 all-player images	[{"images": 831, "source_files": 41510}]	[{"images": 831, "source_files": 41510}]	Yes	3.985

Correctness against the provided source data: **7/7 benchmarks pass**. The benchmark suite covers the historical summary, 2022 standings, player statistics, match results, goal events, player appearances and image descriptors. Reported timings are local observations from this Neo4j instance and vary with hardware, cache state and database load; they are not a cross-system speed claim.

Critical analysis

The graph joins multiple independent sources through explicit match, team, player and provenance edges, reducing the work needed for cross-dataset questions to a single Cypher traversal. The benchmark compares query output with values derived from the input files rather than asking an LLM to judge it. Limitations include a small benchmark set, single-machine timings, and name-based reconciliation when external player identifiers differ across sources. Match-level analysis is limited to fields available in the selected source tables.

6. Conclusions and future work

The implementation provides an end-to-end, reproducible multimedia information retrieval system: source preprocessing and auditing, RDF ontology and triple generation, Neo4j storage, Cypher retrieval, query benchmarks and a Streamlit interface. It includes 2022 match, goal and appearance data, which supports event-level questions rather than only tournament summaries. Player images contribute both descriptive metadata and a simple, inspectable visual feature.

Limitations and future work

The RGB histogram is intentionally simple and captures global color rather than semantics. Cross-source person identity matching depends on normalized names where no common stable key is supplied. The graph does not currently model video clips, tracking coordinates or rich tactical events. Evaluation could be expanded with more held-out questions, completeness checks for every input record, query plans and repeated measurements under controlled cache conditions. Further extensions include richer appearance/event facts, perceptual or learned image descriptors, validated entity-resolution links, indexes and schema constraints.

References and data provenance

1. jfjelstul/worldcup, CSV data at commit 35a8667f518b07469182ae16d35574dd0e7a00fb:
<https://github.com/jfjelstul/worldcup>
2. FIFA World Cup datasets: local professor-provided FIFA 2022 team, player statistics and all-player image files, preserved under dataset/.
3. Cristiano Russo, *Homework: Multimedia Information Retrieval for Sports Analytics*, University of Naples Federico II, A.Y. 2025/2026.